The Toronto Port Lands' Brownfield Site Redevelopment

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Introduction

Since the Industrial Revolution, humans have used a significant amount of land in urban areas for industrial activities. This very often results in the underlying land becoming polluted, which is called a brownfield site (Slawsky et al., 2022). There are an estimated 5 million brownfield sites around the world (Hou et al, 2023), making them a very significant issue that generally does not receive a lot of attention. In Toronto, there is a parcel of land called the Port Lands that was a prime example of a brownfield site. It is located at the mouth of the Don River and the Keating Channel surrounded by Lake Ontario. In this paper, I will argue that the development of the Toronto Port Land's brownfield site will significantly benefit the community by reducing pollution, revitalizing the ecosystem with more habitat space, as well as creating jobs, affordable housing, and more spaces for recreational activities. In doing so, this paper will outline the history of the Toronto Port Land's brownfield site, what redevelopment has been done and the plans for its future, how the project will benefit the local community, why it is beneficial for human heath and the environment to redevelop brownfield sites, how it will relieve housing and job shortages, and my personal opinion about the Toronto Port Lands project.

History of the Toronto Port Lands

Toronto was a trading centre for many centuries due to the access to water routes on Lake Ontario which connects to other lakes and rivers to transport goods. This began long before the first Europeans arrived in the area in the early 17th century, as the First Nations inhabitants used the land that is now Toronto as a hub for trading with other First Nations peoples. Transportation of goods to be sold to people in far away places continued after European colonization. In 1750, French settlers built a trading post called Fort Rouillé beside the shoreline of Lake Ontario. When manufacturing became far more advanced in the 19th century, many factories were built along the shore of Lake Ontario, to provide easier access to shipping the goods they produced. Eventually, the amount of vacant land along the waterfront for additional factories decreased. As a result, the local authorities decided that the land should be artificially expanded into Lake Ontario. This project began in the 1850s and lasted about a century until the 1950s. The area that became the Port Lands was formerly a large marsh called the Ashbridges Bay Marsh before it was artificially filled in to create land for industrial purposes (Waterfront Toronto, n.d.).

Toronto Port Lands Redevelopment

The Toronto Port Lands are located downtown Toronto at the waterfront of Lake Ontario, just to the east of Tommy Thompson Park. A very important feature of the redevelopment is the expanded mouth of the Don River. Since the mouth of the Don River is perpendicular to the Port Lands, it would overflow from storms to flow directly over the Port Lands rather than only through the artificial Keating Channel. The new mouth goes through the Port Lands, which separates part of it from the mainland. This part is a new man-made island called Ookwemin Minising, which has a new park on it named Biidaasige Park. The formerly industrial land has been converted back to nature, with over 5,000 new trees, over 77,000 shrubs, and more than 2 million smaller plants. At the time of writing, most of the Biidaasige Park has been complete, and the incomplete part is expected to be opened in 2026 (City of Toronto, 2025). The park was officially opened by Toronto Mayor Olivia Chow along with several other federal and provincial officials. All three levels of government financially supported the project, as each contributed over $460 million initially for the naturalization part of the redevelopment. Additionally, the federal, provincial, and municipal governments each contributed $325 million to the housing that will be built on Ookwemin Minising, as well as some other development on and around it (City of Toronto, 2025). The Don River overflow has mostly been complete, which will be very important for flood control when the there is significantly more water flowing out of the river after a large amount of rain.

Health Hazards of Brownfield Sites

Brownfield sites have created challenges faced by cities all around the world as they are harmful to both the environment and people's health. Many brownfield sites have the potential to be redeveloped so that they no longer harm people's health and damage the natural environment. Studies have found that people who live close to different brownfield sites disproportionately suffer from lung cancer fatality, higher premature death rates, higher rates of birth defects, and poorer overall health (Wang et al., 2023). This is clearly morally wrong and avoidable, as society could prevent this suffering by redeveloping brownfield sites. Additionally, the many decades of industrial activity on the Toronto Port Lands brownfield site caused the ground to have a very high concentration of petroleum hydrocarbons (Grubb et al., 2025). This has the potential to seriously harm human health, such as cause cancer and developmental issues, even through inhaling the substance (Sahoo, 2024).

Environmental Hazards of Brownfield Sites

It is also unethical how brownfield sites cause harm to the surrounding environment. Most people desire to protect the environment, and redeveloping brownfield sites is one way to accomplish this. A study published in 2021 found that the total petroleum hydrocarbon concentration of the Toronto Port Lands site is about 20,000 mg/kg (Grubb et al., 2025). Petroleum hydrocarbons present in the ground often cause serious damage to surrounding wildlife, especially to aquatic wildlife such as fish. Additionally, petroleum hydrocarbons persist and cause damage for a long time, and bioaccumulate from organism to organism (Sahoo, 2024). Toronto is home to a variety of species of plants and animals, and the natural redevelopment provides plants and animals with habitats, which is very important since there is a shortage of livable space for them in downtown Toronto. Lake Ontario is home to large amount of aquatic wildlife, including the waters around the Port Lands site. Researchers found that environmental restoration projects along the Toronto waterfront have significantly benefited local fish populations, based on evidence over the past few decades from Tommy Thompson Park (Theis et al., 2024), which is beside the Port Lands. This shows that it is possible to improve wildlife populations in the city through brownfield site redevelopment. Additionally, a 2011 study of 90 brownfield sites by the US Environmental Protection Agency found that developing housing in brownfield sites is significantly less harmful to the environment in a variety of ways than developing housing on regular, non-brownfield sites. It concluded that brownfield site housing development produces 32% to 57% less carbon dioxide emissions (Environmental Protection Agency, 2011). Finally, brownfield sites that are not redeveloped present an opportunity cost for reducing greenhouse gas emissions, since redevelopment could give them the potential to be used for the purpose of helping the environment.

Brownfield Sites and the Housing Shortage

The insufficient supply of housing in urban areas is a serious challenge in Canada, as well as in many other developing countries. There are currently significantly more prospective homebuyers and renters than homes available for sale and lease on the market, which naturally causes prices to increase. In Toronto, housing starts have been significantly below the level required for there to be sufficient supply. The federal government of Canada, provincial government of Ontario, and the municipal government of Toronto all have recently implemented policies to increase the supply of housing in their respective jurisdictions, as well as very specific targets for the number of housing units to be built by a target year. The federal government pledges to double rates of homebuilding in Canada by implementing multiple policies, including eliminating the GST for most new home buyers, creating a new entity called “Build Canada Homes” to finance housing construction, reducing development charges by half, and reinstating a tax credit from several decades ago for individuals who invest in constructing rental housing (Sondhi, 2025). These policies could help to significantly increase the supply of affordable housing in Canada, including on redeveloped brownfield sites. In 2023, the Toronto City Council promised to build at least 285,000 homes built by 2031 (City of Toronto, 2023). The Toronto Port Lands Ookwemin Minising housing development is expected to consist of 9,000 housing units by 2031 which will help the federal, provincial, and especially municipal governments in meeting their housing goals (Waterfront Toronto). Government policy on housing is complicated, as there are many different stakeholders involved. It is very important since housing is a necessity and government intervention in the markets is often necessary for sufficient housing supply. It is also very important for people to be able to afford the home they desire because if they cannot then they likely would not be able to have their preferred family formation. A study published in 2025 found that there is a statistically significant negative correlation between housing costs and people living in multi person households in Canada (Lauster & von Bergmann, 2025). Another study that was done in 2023 found that racialized communities in Toronto tend to live in lower income neighbourhoods compared to white people (Geol, 2023), so the new housing on Ookwemin Minising could relieve market pressure and allow for more racialized people to live in higher income neighbourhoods. Additionally, the city of Toronto currently has plans to build a light rail transit from Union Station to Ookwemin Minising (City of Toronto, n.d.). This will make the island even more accessible to both residents and visitors. This is another example of how the public and private sectors are working together to develop the new community on Ookwemin Minising.

Brownfield Sites and Unemployment

Besides a shortage of housing in Toronto, there is also a severe shortage of jobs. The unemployment rate in the city of Toronto in June 2025 was 9.7% (Statistics Canada, 2025). Similarly to how the Port Lands redevelopment will create new housing, the project will create, and has already created, jobs for Torontonians. Even though the number of jobs created will not be sufficient to solve Toronto's unemployment issues, it will make a difference as 3,000 people are expected to work on Ookwemin Minising once it is completed in addition to the large number of people involved in redeveloping the site (Waterfront Toronto). Waterfront Toronto also states that the project, along with some similar redevelopment projects along the Toronto waterfront, will create 100,000 jobs in the skilled trades as well as contribute $13.2 billion to GDP. It is important to remember that each job created very likely represents a significant improvement in an individual person's life, as having a job provides people with income that is required to satisfy their wants and needs. Redeveloping brownfield sites is a very efficient way of transforming economically unproductive land into productive land once again.

Other Brownfield Site Redevelopment Projects

As previously mentioned, brownfield sites are fundamentally harmful and wasteful for urban communities. Similarly to how Toronto is converting the previously industrial land of the Port Lands into housing and parkland, many other cities are also redeveloping their brownfield sites. A project that is similar to the Port Lands redevelopment is taking place in Bristol, England. It is called Brabazon and is being built on the brownfield site of an old airfield. It will have 6,500 housing units, a large park, and a stadium (Swallow, 2025). There are also exciting brownfield site redevelopment projects happening in Canada. An example of one is the Richmond Industrial Centre in Richmond, British Columbia. Unlike Brabazon in Bristol and the Toronto Port Lands, this project will be used for industrial purposes rather than residential and recreational. When completed, it will consist of 12 warehouses that will have 3.2 million square feet of space and be used for distribution purposes. It will also feature sustainable technology such as LED lighting and charging ports for electric vehicles. The brownfield site used to be a landfill for waste from construction projects that started in the 1970s (Shelar, 2025). These two projects, along with the Toronto Port Lands redevelopment, are examples of how cities can transform disused space in urban environments to improve their citizens' quality of life and make the city more modern and liveable for the locals and tourists. Finally, there is a very large development that is proposed at Toronto's Downsview Park which is a brownfield site that was formerly an airport. This redevelopment could provide housing for about 115,000 people (Procter, 2024).

Personal Opinion and Conclusion

In my opinion, the redevelopment of the Toronto Port Lands is a very exciting project that has brought and will continue to bring benefits to the local community for generations. I believe that is a significant step in the right direction for First Nations reconciliation to name the new island and park in the language of the Indigenous peoples of Toronto. I believe this even more strongly since there are not really a lot of prominent places in Toronto have that have names that derive from First Nations languages. There are also other elements of the park that are meant to honour the First Nations peoples, such as dodem animal sculptures and sacred tobacco plants. I personally recently visited Biidaasige Park on Ookwemin Minising and really enjoyed seeing the beautiful green space as well as the site where the housing will be constructed. While this project is ambitious, it is still very likely possible to complete, especially since a significant proportion of it has already been completed. Brownfield sites in Toronto can be successfully redeveloped, as evidenced by the revitalization of the Distillery District (Vitello, 2025). As previously mentioned, Toronto is experiencing a severe housing shortage and redeveloping brownfield sites into new housing will increase the supply to potentially bring prices down. As a young person who worries about being able to afford a home in Toronto in the future, I believe that redeveloping brownfield sites is a part of the solution to building housing that other young people will be able to afford. Overall, I believe that the project is an excellent example of different people and organizations working together to improve Toronto.

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